

CoalPT in BEAST2: Linking the Model to the XML

This guide maps the mathematical model of the Coalescent with Plasmid Transfer (CoalPT; Müller et al. 2025, *PLOS Pathogens*, 10.1371/journal.ppat.1013621) to the parameters and elements in the BEAST2 XML files used to run it.

1 The model

CoalPT models a population of bacteria where each individual carries a chromosome and zero or more plasmids. Going *backward in time*, two types of events can occur:

- **Coalescence**: two lineages merge into a common ancestor.
- **Plasmid transfer**: a plasmid lineage branches off to a different parental lineage (representing a forward-in-time conjugation event).

The result is a phylogenetic *network* $\mathcal{N}$, not a tree. The chromosome tree and each plasmid tree are embedded within this network.

1.1 Rates

Let k be the number of extant lineages at a given time. The **coalescent rate** is

$$\lambda_{\text{coal}} = \frac{k(k-1)}{2 N_e(t)}, \quad (1)$$

where $N_e(t)$ is the effective population size at time t .

For each lineage carrying both the chromosome and n_p plasmids, each plasmid can independently transfer. The **total plasmid transfer rate** across all lineages is

$$\lambda_{\text{transfer}} = \sum_{\ell \in \text{lineages}} \sum_{j \in \text{plasmids}(\ell)} \rho_j, \quad (2)$$

where ρ_j is the per-lineage transfer rate for plasmid j . In the single-plasmid, single-rate case this simplifies to $\lambda_{\text{transfer}} = \rho \cdot m$, where m is the number of lineages carrying the plasmid.

The waiting time to the next event (of either type) is exponentially distributed with rate $\lambda_{\text{coal}} + \lambda_{\text{transfer}}$.

1.2 Joint posterior

The posterior density is

$$P(\mathcal{N}, \boldsymbol{\mu}, N_e, \boldsymbol{\rho} \mid \mathbf{D}) \propto \underbrace{P(\mathbf{D} \mid \mathcal{N}, \boldsymbol{\mu})}_{\text{sequence likelihood}} \cdot \underbrace{P(\mathcal{N} \mid N_e, \boldsymbol{\rho})}_{\text{network prior (CoalPT density)}} \cdot P(\boldsymbol{\mu}) P(N_e) P(\boldsymbol{\rho}), \quad (3)$$

where $\mathbf{D}$ is the set of sequence alignments (one per segment), $\boldsymbol{\mu}$ collects the substitution model and clock parameters, and $\boldsymbol{\rho}$ is the vector of plasmid transfer rates.

1.3 Network density (CoalPT density)

Given a sequence of network events $e_1, e_2, \dots, e_n$ sorted by time, the log-density of the network is

$$\log P(\mathcal{N} \mid N_e, \rho) = \sum_i \left[\underbrace{\log f(e_i)}_{\text{event contribution}} + \underbrace{W(e_{i-1}, e_i)}_{\text{interval contribution}} \right], \quad (4)$$

where, for each event e_i at time t_i :

- If e_i is a **coalescence**: $\log f(e_i) = -\log N_e(t_i)$.
- If e_i is a **plasmid transfer**: $\log f(e_i) = \log \rho_j$, where ρ_j is the rate for the plasmid being transferred.
- If e_i is a **sample**: no contribution.

The interval contribution between consecutive events is the negative of the total hazard integrated over the interval:

$$W(e_{i-1}, e_i) = -\lambda_{\text{transfer}}(e_{i-1}) \cdot (t_i - t_{i-1}) - \frac{k_{i-1}(k_{i-1} - 1)}{2} \int_{t_{i-1}}^{t_i} \frac{1}{N_e(t)} dt, \quad (5)$$

where k_{i-1} is the number of lineages and $\lambda_{\text{transfer}}(e_{i-1})$ is the total transfer rate, both evaluated just after event e_{i-1} .

1.4 Sequence likelihood

Each segment (chromosome or plasmid) has its own embedded tree within the network. The sequence likelihood for segment s is the standard phylogenetic likelihood:

$$P(\mathbf{D}_s \mid T_s, \boldsymbol{\mu}_s) = \prod_{\text{site } j} \sum_{\text{root state } r} \pi_r L_j(r \mid T_s, \boldsymbol{\mu}_s), \quad (6)$$

computed via Felsenstein's pruning algorithm. Here T_s is the tree for segment s extracted from $\mathcal{N}$, π_r are the base frequencies, and L_j is the conditional site likelihood.

2 Example 1: test_coal.xml (simulation + inference)

This XML simulates a network under known parameters, then runs MCMC to recover them. There is no sequence data; only the network density is evaluated.

2.1 Plate diagram

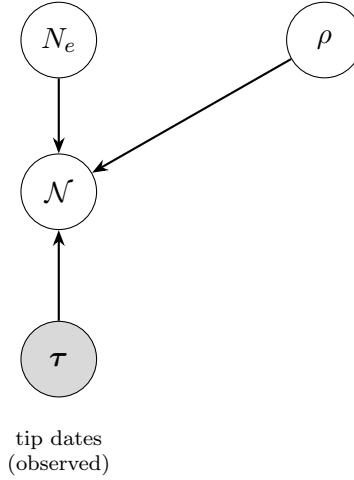


Figure 1: Plate diagram for `test_coal.xml`. Open circles are latent/estimated variables, shaded circles are observed data. In this example the network $\mathcal{N}$ topology is fixed and only N_e and ρ are estimated via MCMC. No sequence data are used.

2.2 Parameter table

Model parameter	Symbol	XML id / attribute	Value
Effective population size	N_e	id="Ne"	estimated (init: 3)
Plasmid transfer rate	ρ	id="transfer"	estimated (init: 3)
True N_e (simulation)	N_e^*	popSize value="0.5"	0.5 (fixed)
True ρ (simulation)	ρ^*	plasmidTransferRate value="5"	5 (fixed)
Number of plasmids	P	nPlasmids="1"	1

Table 1: Parameters in `test_coal.xml`.

2.3 XML-model mapping

Network simulation. The `SimulatedCoalescentWithPlamidsNetwork` element forward-simulates a network under the CoalPT process using the *true* parameter values. The `nPlasmids="1"` attribute means one chromosome + one plasmid (2 segments). The `traitSet` with `date-backward` provides tip sampling times.

```

<!-- The network (simulated under known parameters) -->
<stateNode id="network"
  spec="SimulatedCoalescentWithPlamidsNetwork"
  nPlasmids="1" enableSegmentTreeUpdate="false">
  <populationModel spec="ConstantPopulation">
    <popSize spec="RealParameter" value="0.5"/>
  </populationModel>
  <plasmidTransferRate spec="RealParameter" value="5"/>
  <traitSet ... traitname="date-backward"> ... </traitSet>
</stateNode>

```

Network prior density. The `CoalescentWithPlasmids` element computes $\log P(\mathcal{N} \mid N_e, \rho)$ as described in Section 1.3. The `PlasmidNetworkIntervals` element extracts the sorted event list and computes $\lambda_{\text{transfer}}$ at each event. The parameters `@transfer` and `@Ne` are the *estimated* values (distinct from the simulation truth).

```
<!-- Network prior density: P(N | Ne, rho) -->
<distribution id="coalDensity"
  spec="CoalescentWithPlasmids">
  <networkIntervals spec="PlasmidNetworkIntervals"
    network="@network">
    <plasmidTransferRate idref="transfer"/>
  </networkIntervals>
  <populationModel spec="ConstantPopulation">
    <popSize idref="Ne"/>
  </populationModel>
</distribution>
```

Operators. Only ρ and N_e are sampled; the network topology is *fixed* (no network operators are included). This tests whether the density computation correctly recovers the simulation parameters.

```
<!-- MCMC operators -->
<operator spec="ScaleOperator" parameter="@transfer" />
<operator spec="ScaleOperator" parameter="@Ne" />
```

3 Example 2: Flexneri_equal_rep0.xml (empirical analysis)

This XML performs full Bayesian inference on *S. flexneri* data: chromosome (NC_004337, 31,711 sites) + one plasmid (NC_004851, 1,798 sites). Both the network topology and all parameters are estimated.

3.1 Plate diagram

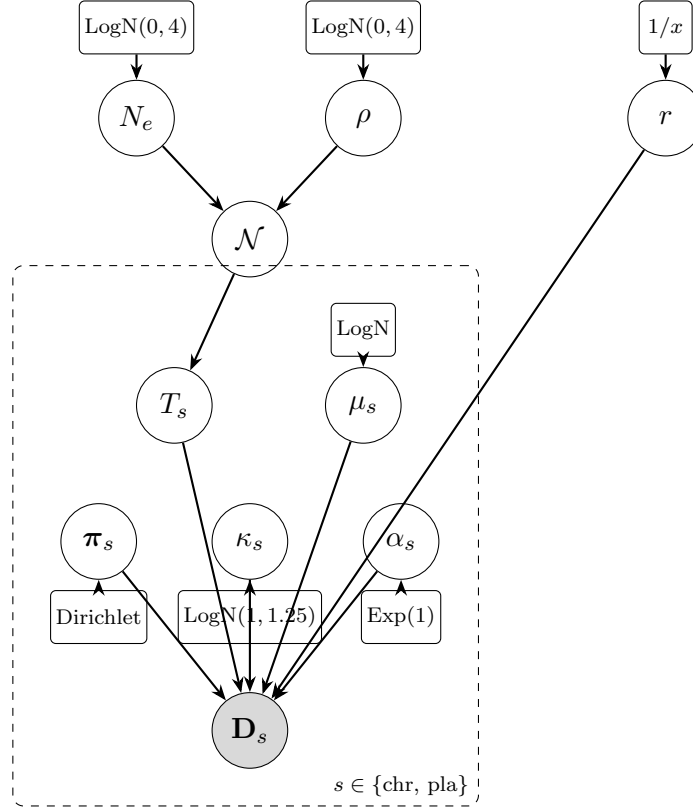


Figure 2: Plate diagram for `Flexneri_equal_rep0.xml`. Rounded rectangles are fixed prior distributions, open circles are latent/estimated variables, and the shaded circle is observed data. The plate (dashed box) is replicated for each segment s (chromosome and plasmid). The tree T_s is deterministically extracted from the network $\mathcal{N}$. The clock rate r is shared across segments; substitution parameters (κ_s , α_s , π_s , μ_s) are segment-specific.

3.2 Parameter table

Model parameter	Symbol	XML id	Initial value
Effective population size	N_e	Ne	100
Plasmid transfer rate	ρ	plasmidTransferRate	0.1
Clock rate (shared)	r	clockRate.c	0.0005
Relative mutation rate (chr)	μ_{chr}	mutationRate.s:NC_004337	1
Relative mutation rate (pla)	μ_{pla}	mutationRate.s:NC_004851	1
HKY κ (chr)	κ_{chr}	kappa.s:NC_004337	1.72
HKY κ (pla)	κ_{pla}	kappa.s:NC_004851	1.53
Gamma shape (chr)	α_{chr}	gammaShape.s:NC_004337	0.50
Gamma shape (pla)	α_{pla}	gammaShape.s:NC_004851	2.73
Base frequencies (chr)	π_{chr}	freqParameter.s:NC_004337	$(0.25)^4$
Base frequencies (pla)	π_{pla}	freqParameter.s:NC_004851	$(0.25)^4$

Table 2: Parameters in `Flexneri_equal_rep0.xml`. The absolute substitution rate for segment s is $r \cdot \mu_s$.

3.3 XML-model mapping

Substitution model. Each segment uses HKY+ Γ_4 with segment-specific parameters:

```
<siteModel spec="SiteModel" gammaCategoryCount="4"
  shape="@gammaShape.s:NC_004337"
  mutationRate="@mutationRate.s:NC_004337">
  <substModel spec="HKY"
    kappa="@kappa.s:NC_004337">
    <frequencies spec="Frequencies"
      frequencies="@freqParameter.s:NC_004337"/>
    </substModel>
  </siteModel>
```

Clock model. A strict clock with a single shared rate:

```
<branchRateModel
  spec="beast...StrictClockModel"
  clock.rate="@clockRate.c"/>
```

A `DeltaExchangeOperator` constrains the relative rates μ_s so their weighted mean equals 1:

```
<operator spec="DeltaExchangeOperator" weight="3.0">
  <parameter idref="mutationRate.s:NC_004337"/>
  <parameter idref="mutationRate.s:NC_004851"/>
  <weightvector spec="parameter.IntegerParameter"
    estimate="false">31711 1798</weightvector>
</operator>
```

Network prior. Same structure as `test_coal.xml`:

```
<distribution id="networkPrior"
  spec="coalpt.distribution.CoalescentWithPlasmids">
  <networkIntervals
    spec="coalpt...PlasmidNetworkIntervals"
    network="@network">
    <plasmidTransferRate
      idref="plasmidTransferRate"/>
    </networkIntervals>
  <populationModel spec="ConstantPopulation">
    <popSize idref="Ne"/>
  </populationModel>
</distribution>
```

3.4 Priors

Parameter	Prior
r (clock rate)	$1/x$ (Jeffreys / OneOnX)
N_e	LogNormal($M = 0$, $S = 4$)
ρ	LogNormal($M = 0$, $S = 4$)
κ_s	LogNormal($M = 1$, $S = 1.25$)
α_s	Exponential(1)
μ_s (relative rate)	LogNormal($M = 0.0005$, $S = 2$, meanInRealSpace)

Table 3: Prior distributions in `Flexneri_equal_rep0.xml`.

3.5 Network topology operators

Unlike `test_coal.xml`, this analysis includes a full suite of operators that modify the network topology:

Operator class	Weight	Description
AddRemovePlasmid	30	Adds or removes a plasmid transfer event
AddRemovePlasmidCoalescent	30	Same, proposing from the coalescent prior
SubPlasmidNetworkSlide	30	Slides a network node up or down
UniformNetworkNodeHeight	20	Uniformly resamples internal node heights
PlasmidNetworkExchange (wide)	10	Wide subtree exchange
PlasmidNetworkExchange (narrow)	5	Narrow (NNI-like) subtree exchange
NetworkScaleOperator	1	Scales all node heights
NetworkScaleOperator (root)	1	Scales only the root height
NetworkScaleOperator (up/down)	1	Scales network+ N_e up, $\rho+r$ down
GibbsOperatorAbovePlasmidRoots	3	Gibbs-samples above plasmid tree roots

Table 4: Network topology operators in `Flexneri_equal_rep0.xml`.

3.6 Inference engine

This XML uses `CoupledMCMC` (parallel tempering with 2 chains) rather than standard MCMC, to improve mixing across the complex network space.

4 Segment indexing convention

CoalPT inherits CoalRe’s segment indexing. Segment 0 is the chromosome; segments $1, 2, \dots, P$ are the plasmids. Each network edge carries a `BitSet` indicating which segments it transmits. For example, with one plasmid:

- `segments={0, 1}`: edge carries both chromosome and plasmid.
- `segments={0}`: edge carries only the chromosome.
- `segments={1}`: edge carries only the plasmid (arises at a transfer node).

A lineage contributes to the transfer rate only if it carries both the chromosome *and* at least one plasmid (i.e., `cardinality > 1`).